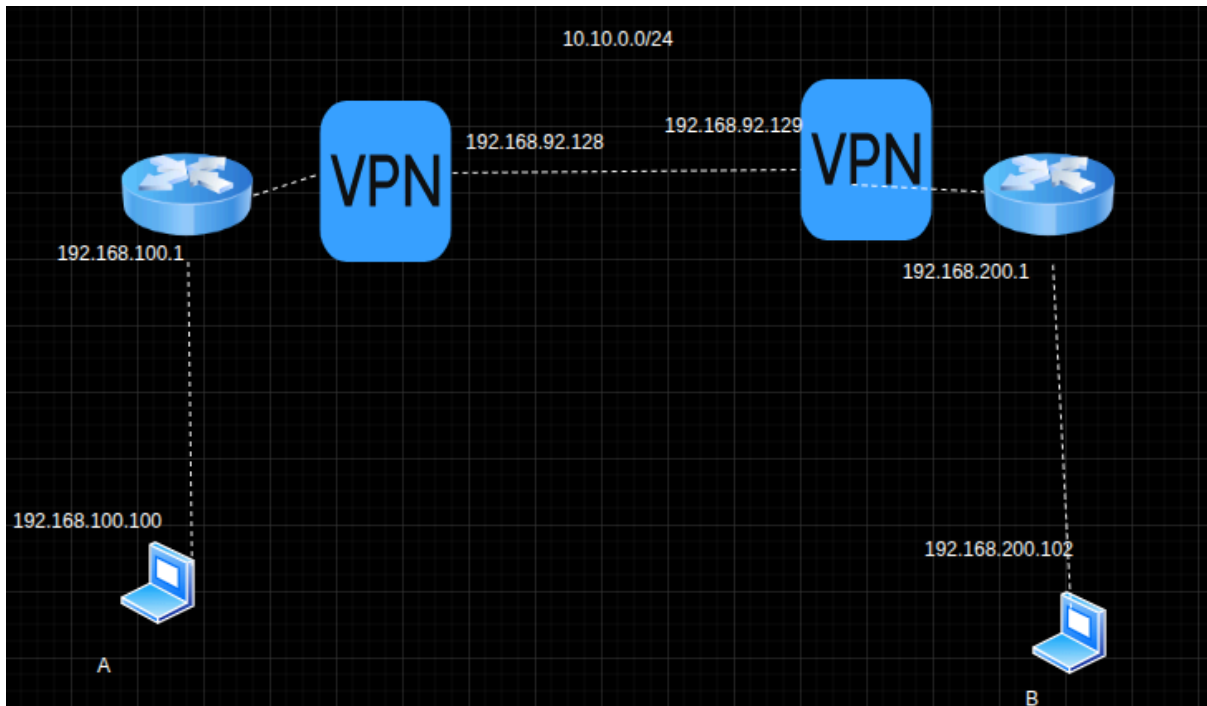
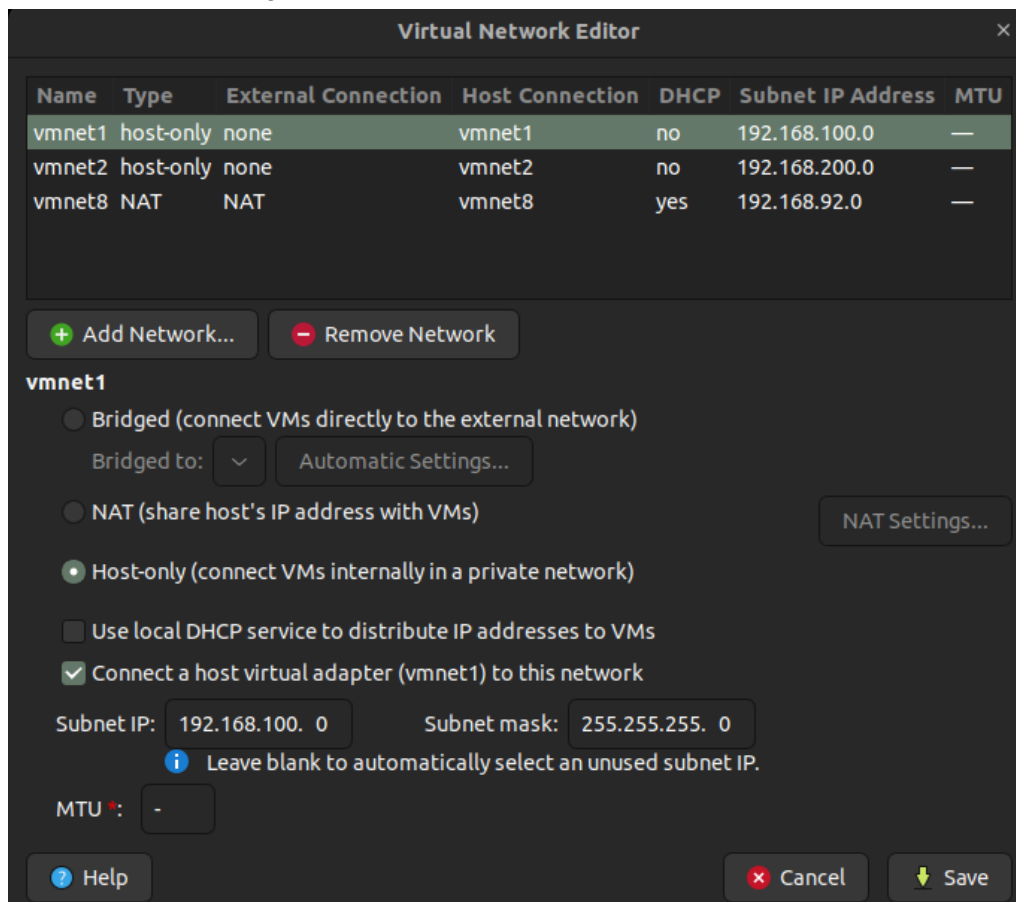




Hình 1. Mô hình mạng

Trong bài lab em thiết lập mô hình Site to Site sử dụng **Share Key**, có cấu hình các máy như sau :

Em thiết lập dải mạng như sau :

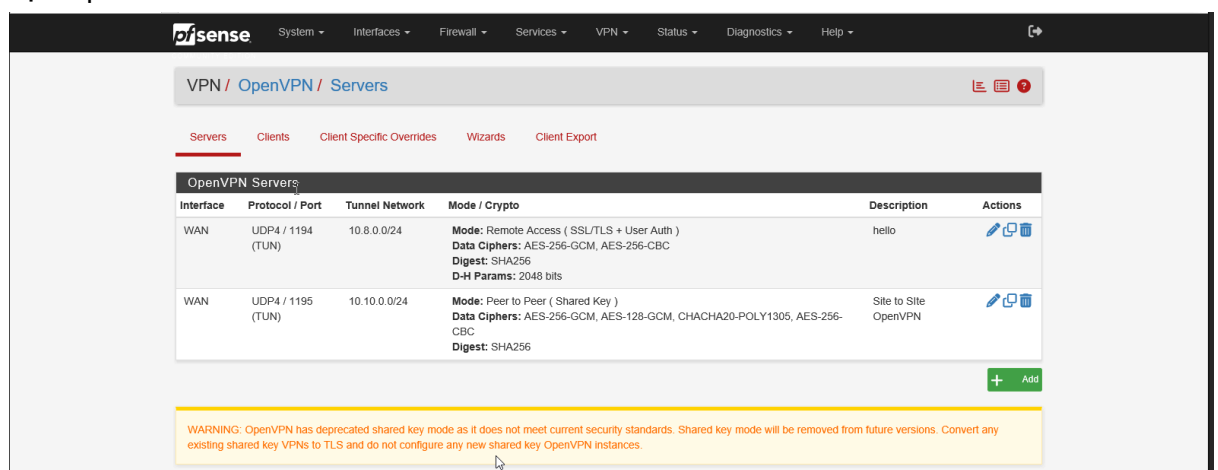


4 máy như sau :

Thành phần	Card mạng trên VMware	IP
Mạng	VMnet8 (NAT)	192.168.92.0/24
A	Card 1 (WAN): VMnet8 (NAT) Card 2 (LAN): VMnet1 (Host-Only)	WAN: DHCP 192.168.92.128(Nhận IP từ NAT) LAN: 192.168.100.1/24
Internal Server A	Card 1: VMnet1 (Host-Only) (DHCP) Nhận IP từ A	IP: 192.168.100.100/24 (Gateway: 192.168.100.1)
B	Card 1 (WAN): VMnet8 (NAT) Card 2 (LAN): VMnet2 (Host-Only)	WAN: DHCP 192.168.92.129 (Nhận IP từ NAT) LAN: 192.168.200.1/24
[Branch] Client	Card 1: VMnet2 (Host-Only) (DHCP)	IP: 192.168.200.102 (Gateway: 192.168.200.1)

A. Cấu hình trên máy A

1. Tạo OpenVPN Server



Server được sử dụng là Site to Site OpenVPN
Cụ thể :

- Description: Site to Site OpenVPN (Đặt tên tùy ý)
- Server Mode: Peer to Peer (Shared Key)
- Protocol: UDP on IPv4 only

- Interface: WAN (interface đích nhận gói tin từ bên ngoài)
- Local Port: 1195
- TLS configuration: Tích chọn ô Automatically generate a shared key. _. Copy Key được sinh ra để Paste vào OpenVPN (Clients) bên chỗ B
- Tunnel Network: 10.10.0.0/24(Đường hầm kết nối giữa 2 router).
- Remote Network: 192.168.200.0/24
- Còn lại để mặc định

Tunnel Settings

IPv4 Tunnel Network

10.10.0.0/24

This is the IPv4 virtual network or network type alias with a single entry used for private communications between this server and client hosts expressed using CIDR notation (e.g. 10.0.8.0/24). The first usable address in the network will be assigned to the server virtual interface. The remaining usable addresses will be assigned to connecting clients.

A tunnel network of /30 or smaller puts OpenVPN into a special peer-to-peer mode which cannot push settings to clients. This mode is not compatible with several options, including Exit Notify, and Inactive.

IPv6 Tunnel Network

This is the IPv6 virtual network or network type alias with a single entry used for private communications between this server and client hosts expressed using CIDR notation (e.g. fe80::/64). The ::1 address in the network will be assigned to the server virtual interface. The remaining addresses will be assigned to connecting clients.

IPv4 Remote network(s)

192.168.200.0/24

IPv4 networks that will be routed through the tunnel, so that a site-to-site VPN can be established without manually changing the routing tables. Expressed as a comma-separated list of one or more CIDR ranges or host/network type aliases. If this is a site-to-site VPN, enter the remote LAN/s here. May be left blank for non site-to-site VPN.

IPv6 Remote network(s)

These are the IPv6 networks that will be routed through the tunnel, so that a site-to-site VPN can be established without manually changing the routing tables. Expressed as a comma-separated list of one or more IP/PREFIX or host/network type aliases. If this is a site-to-site VPN, enter the remote LAN/s here. May be left blank for non site-to-site VPN.

Concurrent connections

Specify the maximum number of clients allowed to concurrently connect to this server.

Allow Compression

Refuse any non-stub compression (Most secure)

Allow compression to be used with this VPN instance. Compression can potentially increase throughput but may allow an attacker to extract secrets if they can control compressed plaintext traversing the VPN (e.g. HTTP). Before enabling compression, consult information about the VORACLE, CRIME, TIME, and BREACH attacks against TLS to decide if the use case for this specific VPN is vulnerable to attack.

Asymmetric compression allows an easier transition when connecting with older peers.

Type-of-Service

☐ Set the TOS IP header value of tunnel packets to match the encapsulated packet value.

2. Thiết lập Firewall

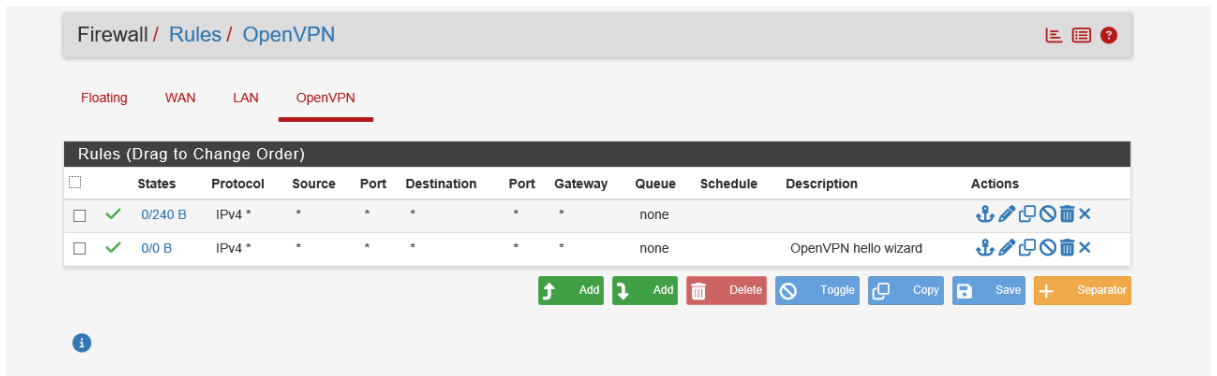
Firewall / Rules / WAN

Floating WAN LAN OpenVPN

Rules (Drag to Change Order)

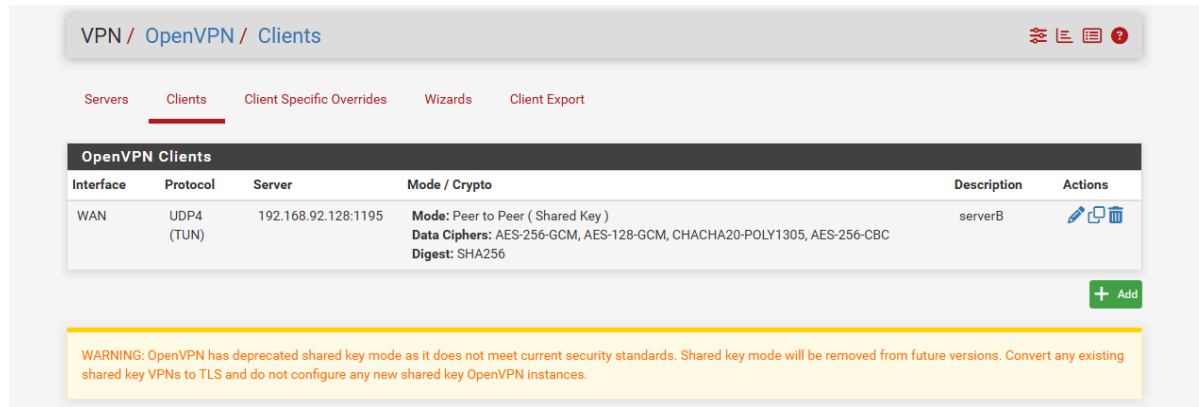
	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/6 KIB	IPv4 *	*	*	*	*	none			    
☐	✓	0/0 B	IPv4 ICMP	*	*	WAN address	*	none			    
☐	✓	0/0 B	IPv4 UDP	*	*	WAN address	1194 (OpenVPN)	none		OpenVPN hello wizard	    

 Add
  Add
  Delete
  Toggle
  Copy
  Save
  Separator

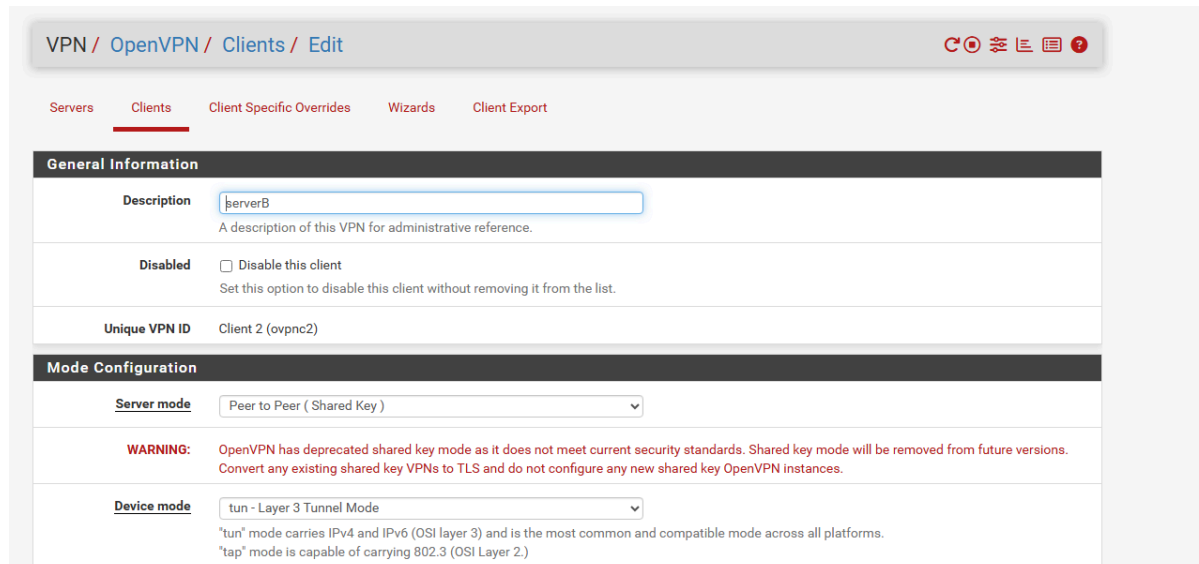


B. Cấu hình trên máy B

1. Tạo OpenVPN (Clients)



Lưu ý : Sử dụng ShareKey được sinh ra ở bên A để Paste vào Use Key



Endpoint Configuration

Protocol

UDP on IPv4 only

Interface

WAN

The interface used by the firewall to originate this OpenVPN client connection

Local port

Set this option to bind to a specific port. Leave this blank or enter 0 for a random dynamic port.

Server host or address

192.168.92.128

The IP address or hostname of the OpenVPN server.

Server port

1195

The port used by the server to receive client connections.

Proxy host or address

The address for an HTTP Proxy this client can use to connect to a remote server.
TCP must be used for the client and server protocol.

Proxy port

Proxy Authentication

none

The type of authentication used by the proxy server.

Cryptographic Settings

Auto generate

☐ Automatically generate a shared key

Shared Key

2048 bit OpenVPN static key

-----BEGIN OpenVPN Static key V1-----
b6efb79eb70d615c1c99f01e31ee2151

Paste the shared key here

Data Encryption Algorithms

AES-128-CBC (128 bit key, 128 bit block)
AES-128-CFB (128 bit key, 128 bit block)
AES-128-CFB1 (128 bit key, 128 bit block)
AES-128-CFB8 (128 bit key, 128 bit block)
AES-128-GCM (128 bit key, 128 bit block)
AES-128-OFB (128 bit key, 128 bit block)
AES-192-CBC (192 bit key, 128 bit block)
AES-192-CFB (192 bit key, 128 bit block)
AES-192-CFB1 (192 bit key, 128 bit block)
AES-192-CFB8 (192 bit key, 128 bit block)

AES-256-GCM
AES-128-GCM
CHACHA20-POLY1305

Available Data Encryption Algorithms
Click to add or remove an algorithm from the list

Allowed Data Encryption Algorithms. Click an algorithm name to remove it from the list

The order of the selected Data Encryption Algorithms is respected by OpenVPN. This list is ignored in Shared Key mode.

Fallback Data Encryption Algorithm

AES-256-CBC (256 bit key, 128 bit block)

The Fallback Data Encryption Algorithm used for data channel packets when communicating with clients that do not support data encryption algorithm negotiation (e.g. Shared Key). This algorithm is automatically included in the Data Encryption Algorithms list.

Auth digest algorithm

SHA256 (256-bit)

The algorithm used to authenticate data channel packets, and control channel packets if a TLS Key is present.
When an AEAD Encryption Algorithm mode is used, such as AES-GCM, this digest is used for the control channel only, not the data channel.
Set this to the same value as the server. While SHA1 is the default for OpenVPN, this algorithm is insecure.

Lưu ý: Bỏ tích chọn: Automatically generate a ShareKey

Tunnel Settings

IPv4 Tunnel Network

10.10.0.0/24

This is the IPv4 virtual network or network type alias with a single entry used for private communications between this client and the server expressed using CIDR notation (e.g. 10.0.8.0/24).

This should be left blank in most cases as servers typically provide addresses to clients dynamically.

The second usable address in this network will be assigned to the client virtual interface. Ensure the Topology setting matches the server when using SSL/TLS and TUN modes or the interface address may not be configured properly. A tunnel network of /30 or smaller puts OpenVPN into a special peer-to-peer mode which cannot receive settings from the server dynamically. This mode is not compatible with several options, including Exit Notify, and Inactive.

IPv6 Tunnel Network

This is the IPv6 virtual network or network alias with a single entry used for private communications between this client and the server expressed using CIDR notation (e.g. fe80::/64). When set static using this field, the ::2 address in the network will be assigned to the client virtual interface. Leave blank if the server is capable of providing addresses to clients.

IPv4 Remote network(s)

192.168.100.0/24

IPv4 networks that will be routed through the tunnel, so that a site-to-site VPN can be established without manually changing the routing tables. Expressed as a comma-separated list of one or more CIDR ranges or host/network type aliases. If this is a site-to-site VPN, enter the remote LAN/s here. May be left blank for non site-to-site VPN.

IPv6 Remote network(s)

These are the IPv6 networks that will be routed through the tunnel, so that a site-to-site VPN can be established without manually changing the routing tables. Expressed as a comma-separated list of one or more IP/PREFIX or host/network type aliases. If this is a site-to-site VPN, enter the remote LAN/s here. May be left blank for non site-to-site VPN.

Limit outgoing bandwidth

Between 100 and 100,000,000 bytes/sec

Maximum outgoing bandwidth for this tunnel. Leave empty for no limit. The input value has to be something between 100 bytes/sec and 100 Mbytes/sec (entered as bytes per second). Not compatible with UDP Fast I/O.

Allow Compression

Refuse any non-stub compression (Most secure)

Allow compression to be used with this VPN instance.
Compression can potentially increase throughput but may allow an attacker to extract secrets if they can control compressed plaintext traversing the VPN (e.g. HTTP). Before enabling compression, consult information about the VORACLE, CRIME, TIME, and BREACH attacks against TLS to decide if the use case for this specific VPN is vulnerable to attack.

Asymmetric compression allows an easier transition when connecting with older peers.

2. Thiết lập Firewall

Firewall / Rules / WAN

Floating WAN LAN OpenVPN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/0 B	IPv4 *	*	*	*	*	none			  
☐	✓	0/0 B	IPv4 *	*	WAN address	*	*	none		OpenVPN hello wizard	  
☐	✓	0/0 B	IPv4 UDP	*	WAN address	1195	*	none			  

 Add  Add  Delete  Toggle  Copy  Save  Separator

Firewall / Rules / OpenVPN

Floating WAN LAN OpenVPN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/240 B	IPv4 *	*	*	*	*	none			  

 Add  Add  Delete  Toggle  Copy  Save  Separator

C. Kiểm tra

1. Trên máy A

Status / OpenVPN

ovpns1: hello UDP4:1194 / Client Connections: 0

Common Name	Real Address	Virtual Address	Last Change	Bytes Sent	Bytes Received	Cipher	Actions

Peer to Peer Server Instance Statistics

Name	Status	Last Change	Virtual Address	Remote Host	Bytes Sent	Bytes Received	Service
ovpns2	Connected (Success)	Mon Jun 1 9:38:00 2026	10.10.0.1	192.168.92.129	4 KiB	4 KiB	
Site to Site OpenVPN UDP4:1195							

```

PS C:\Users\Administrator> ping 192.168.200.102

Pinging 192.168.200.102 with 32 bytes of data:
Reply from 192.168.200.102: bytes=32 time=1ms TTL=126
Reply from 192.168.200.102: bytes=32 time=4ms TTL=126
Reply from 192.168.200.102: bytes=32 time=4ms TTL=126
Reply from 192.168.200.102: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.200.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms
PS C:\Users\Administrator>
  
```

2. Trên máy B

Status / OpenVPN

ovpns1: hello UDP4:1194 / Client Connections: 0

Common Name	Real Address	Virtual Address	Last Change	Bytes Sent	Bytes Received	Cipher	Actions

Client Instance Statistics

Name	Status	Last Change	Local Address	Virtual Address	Remote Host	Bytes Sent	Bytes Received	Service
ovpnc2	Connected	Mon Jun 1 9:44:09 2026	192.168.92.129:47089	10.10.0.2	192.168.92.128:1195	2 KiB	2 KiB	
serverB UDP4								

```

PS C:\Users\Administrator> ping 192.168.100.100

Pinging 192.168.100.100 with 32 bytes of data:
Reply from 192.168.100.100: bytes=32 time=1ms TTL=126
Reply from 192.168.100.100: bytes=32 time=2ms TTL=126
Reply from 192.168.100.100: bytes=32 time=3ms TTL=126
Reply from 192.168.100.100: bytes=32 time=4ms TTL=126

Ping statistics for 192.168.100.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms
PS C:\Users\Administrator>
  
```